

Figure 6: Relative change in water supply risk, under high growth and climate change scenarios, across municipalities when water supply is dependent on local water resources alone (E1) and with interconnected and augmented supply (E2)

Based on current demand projections (driven in part by population growth) and without effective intervention, in addition to the water conservation sub-programmes included in the NWPP, the water deficit confronting the country may range from 2.7 to 3.8 billion cubic meters by 2030, a gap of approximately 17% of available water sources.^{xxxiv} At the municipal level, climate change – coupled with population growth and increased water demand – is likely to result in heightened water supply vulnerability in the majority of municipalities and particularly in the key growth nodes of the country in the Western Cape (Cape Town and linked secondary towns), Eastern Cape (Nelson Mandela Bay), Gauteng (Johannesburg, Pretoria and Ekurhuleni) and Kwa-Zulu Natal (eThekweni). While non-climate stressors such as inefficiencies in governance and suboptimal water infrastructure management (including water losses) do contribute to the challenge, it is climate change-driven variability and decrease in rainfall over several regions of the country that underpin the expected water deficit. Climate change adaptation is thus central to any, and all solutions.

Country Ownership of the Proposed Programme and Alignment with National Priorities

The Department of Water and Sanitation (DWS) has established the National Water Partnerships Programme and Water Partnerships Office (WPO) in the DBSA. The WPO is a ring-fenced entity housed in DBSA. A steering committee chaired by the DG: DWS oversees the work of the WPO. Roles of the WPO include:

- develop standardized national programmes for private sector participation in municipal water and sanitation services, to make it easier, quicker and cheaper for municipalities to enter into partnerships, without having to ‘reinvent the wheel’ for each partnership
- support municipalities to participate in the programmes and preparation of projects – where municipalities are lacking in the required expertise and funding to undertake bankable feasibility studies and financial structuring
- where appropriate, facilitate blended financing, including participation by DFIs.

The WPO has received seed funding from NT and DWS has requested a budget for the WPO. The WPO structure includes a head of office and a programme head for each of the standard programmes, in addition to other resources that will be added over time. Posts in the WPO are currently being established and will be filled shortly. In the interim the WPO is being run by existing DBSA officials.

The WPO has identified the following standardized programmes (other programmes, such as an Off-grid / non-sewered sanitation programme may be added in future) presented in Table 2.

Table 2: Programmes to be initiated under the NWPP

No	Programme	Description
1	Non-revenue water	Comprehensive water conservation and water demand management and cost recovery programme focusing on reducing losses, reducing over-consumption and improving cost recovery. Private sector obtains return on investment from savings through reducing losses or from increases in revenue
2	Management contract	Programme in which private sector provides overall management support to W&S function in municipality, including both engineering and non-engineering (e.g. revenue collection) aspects. Private sector returns funded from savings through efficiency improvements or from increased revenue
3	Wastewater treatment	A programme focusing on assisting municipalities to upgrade, refurbish and rehabilitate their waste water treatment facilities. Funding and implementation will be done through the PSP Model which leverages a portion of future grant funding
4	Water re-use	Programme focusing on further treatment of municipal wastewater to enable it to be resold for either potable, industrial or agricultural use. Related resource recovery includes energy generation and sludge beneficiation. Private sector obtains return on investment from sale of re-used water, generation of energy and sludge beneficiation
5	Sea water desalination	Focus on independent water production by producing potable water through seawater desalination in coastal cities. Private sector obtains return on investment from sale of desalinated water to municipalities or other customers (with the approval of the WSA)

To prepare for a hotter and drier future with dwindling freshwater resources and growing water demand, South Africa developed its National Water and Sanitation Master Plan in 2018. The Master Plan (which built on an earlier National Strategy for Water Re-use)^{xxxv} identifies water re-use as one of the central strategies in its overall approach to ensuring water security and climate resilience.

The National Climate Change Adaptation Strategy (NCCAS) provides for a clustered approach that outlines a range of interventions to increase human resilience, economic resilience, environmental and ecological infrastructure resilience, and physical infrastructure resilience. It recognizes the social and technical capacity barriers, the institutional challenges and fragmentation in approach, the need for new technologies as well as the key financial constraints that inhibit adaptive response. Implementing water re-use nationwide does support the major tenets of NCCAS, which urges water management institutions to adopt more adaptive responses (action 1.1.28) and underscores the need for more water-wise water management practices in urban areas – including water re-use (action 1.1.32).^{xxxvi}

South Africa's updated draft Nationally Determined Contribution (NDC)^{xxxvii} reinforces this commitment further through its Goal 3: implementation of the NCCAS interventions for the period 2021-2030, including in the priority sector of water.^{xxxviii} The government has communicated to the United Nations Framework Convention on Climate Change (UNFCCC), and the broader global community, that water re-use is amongst its highest climate change adaptation priorities by showcasing the National Water Conservation and Water Demand Management Flagship Programme (within which the water re-use programme is situated) in South Africa's most recent Biennial Update Report (BUR).^{xxxix} Water re-use is also consistent with South Africa's National Development Plan (NDP), applicable through 2030, as it aligns with the NDP's water access and water security targets (such as ensuring all South Africans have access to clean running water in their homes, ensuring that there is enough water for agriculture and industry, and developing a national water conservation programme to improve water use and efficiency).^{xl}

The NCCAS was developed in 2019 and is aligned with the Paris Agreement's Article 7 and the associated aspects of the Paris Rulebook. The Strategy further outlines nine strategic objectives to which sectoral responses need to be aligned. The NCCAS, therefore, is the key domestic policy instrument to guide implementation, and has informed an updated adaptation communication as a component of its Nationally Determined Contribution (A-NDC). The A-NDC has outlined a number of relevant goals and associated actions as tabulated in Table .

Table 3: South Africa's First Adaptation Communication^{xli}

Elements	Undertaking for the period 2021-2030	Assumptions / Methodologies / Context	Efforts	Adaptation investment (2021 - 2030)
Element (c) of the Annex to decision 9/CMA.1 National adaptation priorities, strategies, plans, goals and actions	Goal 3: Implementation of NCCAS adaptation interventions for the period 2021 to 2030	The local government plays a key role in climate change response, and therefore building the capacity of the local sphere of government will be significant in achieving adaptation goals. This capacity support should be inclusive of human resources; institutionalisation of climate change response; financial resources and technological and/or technical support. The cities will play a pivotal role in leading climate change response in the country by virtue of urbanisation trends and services offered to the community.	The NDC will prioritise specific sectors and work towards the development of climate response plans to address the specific sector priorities. The priority sectors are health, water, biodiversity, agriculture and human settlements. Water sector: enhance water security; effectively deploy flood protection measures, and hydro-metrological monitoring systems. Infrastructure: Ensure the development and deployment of climate-resilient infrastructure that enhance water and energy security. Integration of climate information into infrastructure development planning.	USD 3 - 4 billion required for the implementation of the NCCAS for a period 2021 to 2030 to support South Africa.
Element (d) of the Annex to decision 9/CMA.1 Implementation and support needs of, and provision of adaptation support to South Africa	Goal 4: Access to funding for adaptation implementation through multilateral funding mechanisms	The 2009 UNFCCC report estimates that developing countries such as South Africa will require between USD \$23-67 billion by 2030 to adapt to climate change. The domestic financial sector should play a pivotal role in terms of helping investors in adaptation space to satisfy funding requirements to meet the NDC goals. Direct unilateral access to advance adaptation finance by the private sector is still a major issue. The National Treasury, South African Reserve Bank, financial sector regulators and Department of Forestry and Environment (DFFE) work collaboratively on issues of Sustainable finance initiatives in the sphere of private finance.	Development of climate change adaptation investment pipeline projects.	Adaptation needs and costs for the period 2021 – 2030 is USD 16 – USD 267 billion. Adaptation needs and costs adapted by a minimum of 4% GDP impact is USD 122 billion by 2025 and USD 37 5 billion by 2030.

From a climate change adaptation perspective, water re-use is already a proven resilience strategy in South Africa after being successfully deployed as a response measure^{xliii} by the CCT in the aftermath of the 2018 “Day Zero” drought – a climate-driven event that saw a city of nearly five million run out of water (attribution studies^{xliiii} have established that beyond the probability linked to natural climate variability, the drought was made between three^{xliv} to six^{xlv} times as likely^{xlvi} by climate change). During the crisis, the City commissioned a water re-use plant to provide 10 million litres a day.^{xlvii} The City already sells treated wastewater for irrigation, industrial, and construction purposes, and would be well-placed to increase reuse for potable and other non-potable water use.^{xlviii} Other water-stressed cities in the country must follow suit.

For the cost-effectiveness and sustainability of water re-use as a climate change adaptation measure, it is important that cities in South Africa mainstream such approaches. As a recent study noted, “prior to the drought, seawater desalination and water reuse ranked lower in the CCT’s water reconciliation strategies –

only to be accorded priority during the drought period when the City experienced severe water shortages. These options should not be considered crisis options, but rather be considered as normal options at all times. At such instances, these projects are hastily considered and implemented usually at a high premium. Such an approach is not cost efficient, resulting in end-users paying exorbitant price for the water.^{xlix}

It is with this intent, i.e. to establish and scale-up water re-use infrastructure as the norm where there is climate induced water insecurity in municipalities, that South Africa has designed the proposed national water re-use programme.

It is important to consider the impacts of climate change in the context of current vulnerability as well as future resilience to climate shocks and the potential for adaptation. The Green Book risk assessmentⁱ (CSIR and Aurecon, 2019) has provided the most thorough assessment of national and municipal vulnerability in South Africa to date, being completed in 2019 and being updated regularly. This assessment has shown by looking at climate change impact in conjunction with water supply contexts (local supply E1 and regional supply E2), against differing population growth scenarios that there is significant vulnerability across many municipalities. By taking a median climate scenario and looking at the adjustments in vulnerability from the current context to futures with medium and high population growth, these vulnerabilities often increase between 20—40% and in some instances more. This particularly in key country economic nodes of Gauteng, Kwa-Zulu-Natal, Eastern Cape and Western Cape, where rapid urbanisation around the cities of Johannesburg/ Tshwane, eThekweni, Nelson Mandela Bay and Cape Town are taking place.

In looking at possible pathfinder projects to be undertaken by the WRP, a longer-term pipeline was shortlisted to ten indicative projects across the country. While this assessment is not exhaustive it provides a useful indication of potential pipeline for embedding the programme. Using these localities, an assessment of vulnerabilities was undertaken using the Green Book municipal vulnerability assessment tool as well as the WRI Aquaduct Risk Toolⁱⁱ for water stress by the year 2040.

The vulnerabilities at municipal level are summarised in Table . While all show increased risk and vulnerability due to the impacts of climate change a number of these do not demonstrate strong climate arguments. Thus, in preparing the project pipeline for the WRP the need for more detailed vulnerability assessments is imperative and will be undertaken as a first step prior to projects being provided project preparation support. Only projects with clear climate arguments will receive financial support from the WRP and this has been incorporated into the project selection criteria.

The Environmental and Social Management Framework for the WRP outline the importance of ensuring that all projects adequately address all related environmental and social risk. These detailed vulnerability studies must address these risks.

Table 4: Summary of municipal vulnerabilities for potential projects

PILOT MUNICIPALITIES (Showing Treated Water Output from WRP Ml/d)	Green Book Projected Vulnerability 2050 Median Climate Scenario Exposure E2		WRI Aquaduct Risk Tool
	Medium Population Growth	High Population Growth	Water Stress 2040
Eastern Cape			
Nelson Mandela Metropolitan Municipality (40 Ml)	40-60% deficit	60-80% deficit	>80% Extremely High
Free State			
Mangaung Metropolitan Municipality (25 Ml)	0-20% surplus	0 -20% deficit	20-40% Medium High
Gauteng			

City of Ekurhuleni (60 MI)	40-60% deficit	60-80% deficit	20-40% Medium High
City of Johannesburg Metropolitan Municipality (50 MI)	60-80% deficit	100% deficit	20-40% Medium High
City of Tshwane Metropolitan Municipality (30 MI)	40-60% deficit	60-80% deficit	>80% Extremely High
Kwa Zulu Natal			
eThekweni Metropolitan Municipality (100 MI)	20-40% deficit	40-60% deficit	20-40% Medium High
Western Cape			
City of Cape Town Metropolitan Municipality (40 MI)	20 -40% deficit	40-60% deficit	>80% Extremely High
Drakenstein Local Municipality (10 MI)	0 - 20% surplus	0 – 20% deficit	>80% Extremely High
Indicative Projects			
Indicative Project A	20 -40% deficit	40-60% deficit	>80% Extremely High
Indicative Project B	40-60% deficit	60-80% deficit	>80% Extremely High

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

There is a growing realization that the South African water sector needs progressive and innovative approaches to enable adaptation to climate change impacts and ensure a resilient and water secure future. The provision of local water services to communities and business is the responsibility of municipal government who are required to initiate and undertake technical projects such as water reuse projects. This programme will transform the provision of these water services while building resilience to climatic variabilities and the adverse effects of climate change. In doing so it will strengthen the technical and financial capacity of these institutions as well as the broader water sector.

This is aligned to the 2018 NWSMP, the 2019 National Climate Change Adaptation Strategy and the 2021 First Nationally Determined Contribution that make clear statements regarding the importance of addressing a range of systemic issues, undertaking infrastructural interventions as well as looking to ecosystem-based approaches in a structured manner to revolutionize urban water systems and ensure sustainable adaptation to climate change related water security challenges.

Therefore,

If technical and financial support is provided to municipalities in the identification, preparation and implementation of water reuse projects for climate adaptation at scale through a pipeline of projects, **Then** the local socio-economic resilience of communities to climate change will be strengthened through improved water security, **Because** transformational change in public and private sector funding of water reuse infrastructure will enable the establishment of a new asset class that will underpin the upscaling of water reuse nationally. .

The WRP will therefore directly reduce the climate vulnerability of communities through the scaling-up of water reuse projects at municipal scale. These infrastructure projects will be enabled through the introduction of programmatic support to municipalities to develop prepare and implement these projects as well as by developing and providing innovative public-private blended financing arrangements. This will not only